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PHYSICS

Ans 1. Highest resistance can take place in series combination $\Rightarrow R_{eq} = R_1 + R_2 + R_3 + R_4 \dots$
 $= 4 + 8 + 12 + 24$
 $= 48 \Omega$

Lowest resistance can take place in parallel combination $\Rightarrow \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$
 $= \frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{24}$
 $= \frac{12}{24}$
 $= \frac{1}{2}$

$$R_{eq} = 2 \Omega$$

Ans 2. $V = 12 V$
 $I = 0.0025$

So,

$$V = IR$$

$$R = \frac{V}{I}$$

$$\Rightarrow \frac{12}{25} \times 10^4 \times 100$$

$$\Rightarrow 48 \times 100$$

$$\Rightarrow 4800 \Omega$$

Ans 3.

$$\rho = 1.7 \times 10^{-8} \Omega \text{ m}$$

$$\text{Area} = 2 \text{ mm}^2$$

$$= 2 \times 10^{-6} \text{ m}^2$$

$$l = 1 \text{ m}$$

We know,

$$\rho = \frac{RA}{l}$$

$$R = \frac{\rho l}{A}$$

$$= \frac{1.7 \times 10^{-8} \times 1}{10 \times 2 \times 10^{-6}}$$

$$= 8.5 \Omega$$

Ans 4. Electric potential (equivalent) = $V_1 + V_2$
 $= 10 + 8$
 $= 18 \text{ V}$

$$\Rightarrow I = \frac{V}{R}$$
$$= \frac{10}{5} = 2 \text{ A}$$

$\Rightarrow$ And,

$$\frac{R}{4} = \frac{10}{8}$$

$$R = \frac{10^5 \times 4}{8}$$

$$\boxed{R = 5 \Omega}$$